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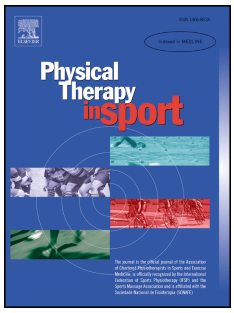

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Secondary injury prevention reduces hamstring strain and time-loss groin injury burdens in male professional football

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Abstract

Objectives: This study aimed to assess the effectiveness of a systems-based secondary injury prevention intervention aimed at early detection and management of hamstring strain injury (HSI) and time-loss groin injury in an Australian male professional football club.

Design: Prospective cohort study.

Setting: Australian male professional football club.

Participants: Data were collected from male professional football players (*n*=73) from a single football club.

Main Outcome Measures: Unilateral knee flexion and side-lying hip adduction maximum voluntary isometric contractions were monitored routinely in-season, two days post-match (≥40 hours [h]) during a three-season intervention period. Strength reductions greater than the tests' minimal detectable change percentage prompted intervention. HSI and time-loss groin injury burdens were calculated per 1000 player hours and compared with those from an immediately preceding two-season control period, to assess the effectiveness of the intervention.

Results: Across the intervention period, there was a decrease in HSI (4.98 days absence/1000h (19.8%) decrease) and time-loss groin injury burdens (0.57 days absence/1000h (49.1%) decrease) when compared with the control period.

Conclusions: A systems-based secondary injury prevention intervention shows preliminary positive findings in reducing HSI and time-loss groin injury burdens within a male professional football club, compared with usual care only.

Keywords: Sports medicine; screening; injury surveillance; team sport; soccer

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Introduction

Hamstring strain injury (HSI) and time-loss groin injuries are common injuries in Australian male professional football, accounting for between 12% to 22% and 8% to 18% of match-loss injuries, respectively.[1] In male professional football HSI incidence and burden rates have increased annually between 2001 and 2014,[2] while time-loss groin injury burden has remained consistent over a similar period (2001 – 2016).[3] Currently, evidence supporting the effectiveness of primary prevention exercise programs in preventing muscle injury in elite football players is limited, with a deterioration of compliance over the season shown previously in football studies.[4,5] As primary prevention exercise programs alone have not been shown to significantly reduce these injuries,[4,5] a complementary systems-based approach, incorporating usual primary and tertiary injury prevention care alongside suitable secondary injury prevention for the early detection and management of HSI and time-loss groin injuries may be more appropriate.[6,7]

Secondary injury prevention involves screening for subclinical signs or symptoms in the period between early detection through clinical screening and the usual time of diagnosis[8] following an exposure to a particular stressor (such as high intensity workloads in match-play football). It aims to allow early intervention or treatment to prevent progression or worsening of the injury.[9] Isometric strength reductions may precede the onset of HSI,[10] while uninjured male professional football players demonstrate stable hip strength capacity in-season.[11] Regular monitoring of in-season isometric strength may therefore form the basis of secondary injury prevention as part of a wider complementary systems-based approach.

Football match-play can cause muscle damage and incite acute inflammatory responses[12] and strength deficits,[13] all of which may inform the clinical reasoning and injury risk management process. Unilateral knee flexion maximum voluntary isometric contraction (MVIC) peak force is significantly reduced immediately and 24 hours (h) post-match before returning to pre-match levels 48 h post-match in elite youth footballers.[13] Lingering strength deficits following this period may indicate incomplete recovery that should be considered for follow-up by sports medicine staff before exposure to further workloads.

Previous research investigating the value of a systems-based approach, whereby usual primary and tertiary injury prevention care is complemented with secondary injury prevention, shows promise in reducing HSI incidence and burden[6] and hip and groin injury burden.[7] In this research HSI and groin injury burden were able to be reduced via a strength screening alert and early indicated intervention system in male youth football,[6,7] however the feasibility and effectiveness of secondary injury prevention of HSI and time-loss groin injuries in male professional football over multiple seasons, is unknown.

The purpose of this study was to assess the effectiveness of a systems-based secondary injury prevention intervention for the early detection and management of HSI and time-loss groin injuries in an Australian male professional football club. It was hypothesised secondary injury prevention coupled with usual primary and tertiary injury prevention care would reduce HSI and time-loss groin injury burdens.

Methods

Study design and procedure

We conducted a prospective cohort study over five seasons in a cohort of professional football players. The first two seasons served as the control period (2016/17 and 2017/18 seasons), while the following three seasons served as the intervention period (2018/19, 2019/20 and 2020/21 seasons). Data were collected from male football players (n = 73) from a single football club. All players from the men's team were included in the study with players joining or leaving the team mid-season followed for their respective periods. As such, the number of players varied over the five-season study period (21 in 2016/17, 25 in 2017/18, 28 in 2018/19, 29 in 2019/20 and 31 in 2020/21). Similarly due to the longitudinal study design, there were four head coach, two physiotherapist and two strength and conditioning coach changes across our five-season monitoring period. Ethical approval for the research was obtained (protocol number H-2018-0229) and all players provided informed consent to participate in the study.

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For the first three seasons of this prospective cohort study (2016/17 – 2018/19), pre-season began in July with the in-season commencing in October and ending in May. However, the COVID-19 pandemic meant there was a three-month league-wide postponement of all football-related activities from March to June in season 2019/20. Similarly, schedule changes arising from the pandemic meant a shortened pre-season for the 2020/21 season, with pre-season beginning in October and in-season in December.

Control period methodology

Usual care was provided to players in the control period in the form of primary and tertiary injury prevention. Players underwent fatigue monitoring testing in the morning prior to every field training session to ascertain training status, namely hip adduction strength and hamstring flexibility testing. These tests were not classified as part of a secondary injury prevention program due to a lack of test sensitivity in detecting subclinical injury and were merely used as fatigue monitoring measures. Hip adduction strength was assessed bilaterally with a standard commercially available aneroid sphygmomanometer (Cumper and Robbins, Australia Medical Diagnostic Equipment Supplier) via the adductor squeeze test in short and long-lever test positions. For short-lever assessment the device was placed between the knees while in a supine position with 45° of hip flexion, while for long-lever assessment the device was placed between the medial malleoli while in a supine position with 0° of hip flexion and the feet placed flat on the floor.[14] Strength was measured via pressure exerted (in millimetres of mercury (mmHg)) with the sphygmomanometer recognised as a reliable and valid assessment of isometric hip strength in athletic populations.[15,16] Hamstring flexibility was assessed via the sit-and-reach test with the result quantified in centimetres (cm). The sit-and-reach test is a reliable and valid method of assessing hamstring flexibility and its protocol has been described previously.[17] A discrepancy of greater than one standard deviation from the player's seasonal mean was used as an alert for further assessment with the resultant procedure outlined in Supplementary Table 1. Management of player training status was assessed by the team strength and conditioning coach, physiotherapist, and sport scientist on a case-by-case basis. Indicated interventions to field training sessions were proportionally based in response to player presentations to daily fatigue screening outlined above, as well as subjective well-being questionnaire and workload data. These included the reduction or removal

of participation in drills containing high-speed running exposure, high-intensity acceleration and deceleration and crossing.

Intervention period methodology

Usual care was provided to players in the intervention period in the form of primary and tertiary injury prevention. In addition to usual care, a secondary injury prevention system was incorporated for the early detection of subclinical signs or symptoms. Testing was administered in the morning prior to the first training session of the week (two days post-match (MD+2) and following a rest day (≥40 h)). Players were tested for unilateral knee flexion (prone, 45° hip flexion and 30° knee flexion) and side-lying hip adduction MVIC peak force (in Newtons [N]) using an externally fixed dynamometer (Applied Measurement Australia Pty Ltd, Melbourne, Australia). Isometric knee flexion testing was performed in accordance with previously published research to capture unilateral MVIC peak force, whereby players exerted a five second MVIC against a dynamometer externally fixed to the floor directly underneath the physiotherapy plinth.[18] A side-lying unilateral isometric hip adduction testing protocol was also integrated.[19] Intra-rater reliability of the protocols was investigated via 29 uninjured players of the intervention period based on a single-measurement, absolute-agreement, two-way mixed-effects model. Calculations indicated this sample size was sufficient, based on minimal acceptable reliability $\rho_0 = 0.7$, expected reliability $\rho_1 = 0.9$, $\alpha = 0.05$ and $\beta = 0.20$.[20] Intraclass correlation coefficient ($ICC_{3,1}$) estimates for isometric knee flexion and hip adduction were 0.83 – 0.90 and 0.66 – 0.76, respectively, with minimal detectable change percentages (MDC%) 19.02% – 23% and 32.98% – 34%, respectively.

An alert was created if a player had isometric knee flexion and/or hip adduction strength reductions exceeding those of the respective test's MDC values, when compared with the players' 8-point moving average (most recent 8 tests). Up to four alerts per player per testing session could be triggered from each respective test and limb combination (left and right-sided unilateral knee flexion and hip adduction). As MD+2 training sessions were very short in duration (< 45 minutes) and prescribed at low intensities to facilitate recovery from the recent match, any alert generated on MD+2 did not affect

a player's inclusion in this session unless otherwise agreed upon with medical staff. Re-testing then occurred in the morning prior to training on MD+3, providing an additional 24 hours for the player to recover from the reduction in strength. Only when strength deficits were still not restored on MD+3 in comparison to their 8-point moving average did they then progress to the remaining stages of the secondary injury prevention system (as per Figure 1). Where indicated, and following clinical examination, subsequent interventions included daily physiotherapy treatment, inclusive of manual therapy, prior to field training sessions for the remainder of the training week. Individualised muscle activation, and strength programs were also to be completed before field sessions. These consisted primarily of the Copenhagen adduction exercise, isometric adduction, eccentric adduction sliders, banded single- and double-leg bridging, lateral band walks, hop/jump landing practice and balance tasks on uneven surfaces with/without volleys (Supplementary Table 2). Load management during field sessions focussed on a reduction in risk exposures, namely high-speed running efforts and distance, explosive change of direction, and repetitive crossing and shooting actions.[21] A shift to an on-field playing position during training which inherently requires execution of less high-intensity activities, such as a shift from wide to central positions,[22] a reduction in the number of involvements in repetition-based drills, or a complete removal from the drill were specific indicated interventions. Consultative decision-making between the sports medicine staff and head coach within the context of the planned training session was key across the intervention period. Post-training recovery modalities were encouraged, inclusive of hydrotherapy and/or ice bath immersion.

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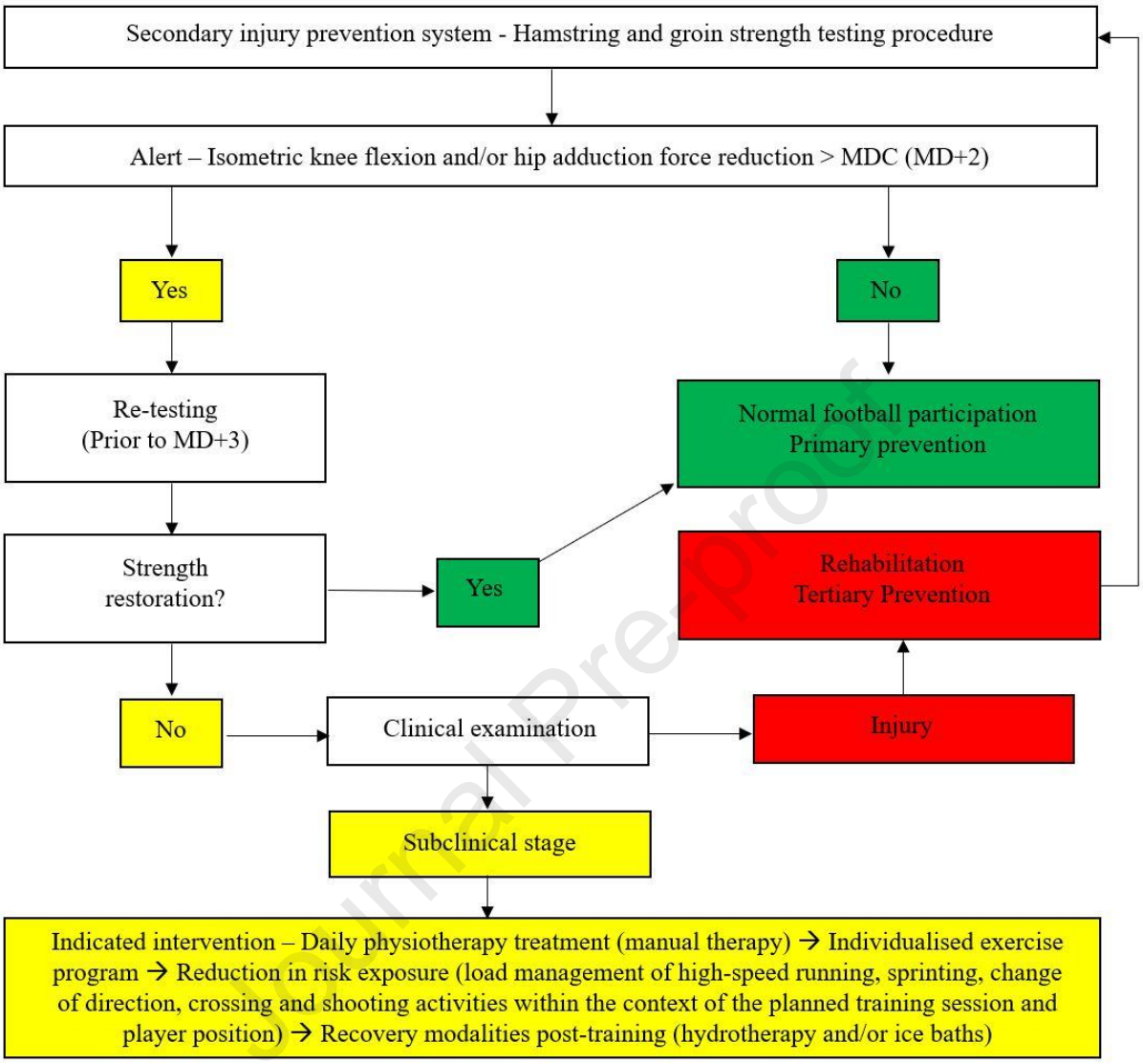

Figure 1 (online version): Secondary injury prevention system for HSI and time-loss groin injuries (adapted from Wollin et al [6,7])

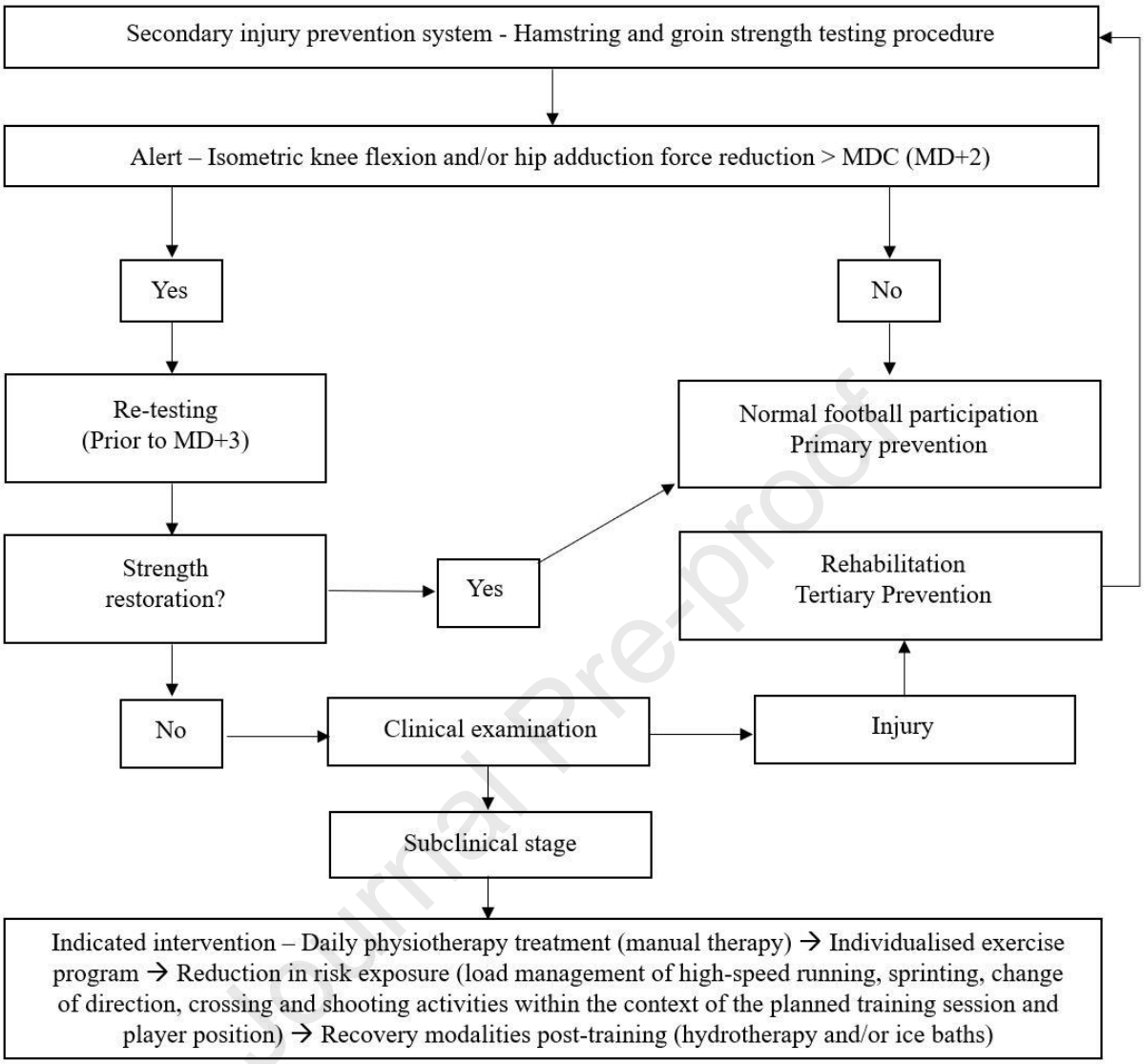

Figure 1 (print version): Secondary injury prevention system for HSI and time-loss groin injuries (adapted from Wollin et al [6,7])

Player workload management

Player workload was individually monitored throughout the season by the team strength and conditioning coach and sport scientist, with a standardised monitoring protocol utilised across both reporting periods. All monitoring data were recorded using password protected athlete monitoring software (Athlete Monitoring, Moncton, Canada), with player training status reviewed daily across both reporting periods. Workload was quantified using both internal and external load measures and was collected for all field and strength training sessions (inclusive of recovery/rehabilitation sessions) and match-play (friendly and competitive). Global positioning system (GPS) data was used to quantify external load measures across all field sessions with monitored variables including total distance, and distances covered at specific thresholds, namely high-speed distance (>19.8 $km.h^{-1}$), and sprinting distance (>25.2 $km.h^{-1}$). Internal load was collected across all sessions and was quantified by the session rating of perceived exertion (sRPE) method[23] which has been validated in football.[24] In addition, players across both reporting periods completed a club-tailored daily subjective well-being questionnaire assessing sleep quality, muscle soreness and fatigue every morning to better inform training status. In-season, match-play typically occurred once per week occurring on a Friday, Saturday, or Sunday.

Injury definition and data collection

An injury was defined as *any physical complaint resulting from a football match or training leading to the player being unable to fully participate in future matches or training* (time-loss definition).[25] As such, both acute and longstanding groin injuries were encompassed within this definition. All injury assessments were performed by the team physiotherapist and/or sports physician and a player was considered injured until they were cleared to be available for match selection. Injuries were categorised by severity and classified as being either: minimal (1 - 3 days absence from training and/or match-play), mild (4 – 7 days absence), moderate (8 – 28 days absence) or severe (greater than 28 days absence).[26] The surveillance period for the study included both the pre-season and in-season, inclusive of all matches and training sessions. If a player was injured at the end of the season, a return-to-play date was estimated based on all available evidence.[25] All data collection protocols and data reporting adhered to

the consensus statement on injury definitions and data collection procedures in studies of football injuries.[25]

Statistical analysis

Normality of the data was assessed using the Shapiro-Wilk test and visually via Q-Q plots. Welch's *t*-test for independent samples, equal variances not assumed, was used for parametric data (age, height, and mass), while Mann-Whitney *U* test was used for non-parametric data (weekly training sessions, weekly matches) to investigate differences between the control and intervention periods.[27] Effect sizes were calculated using Hedges *g* for parametric data and the rank-biserial correlation *r* for non-parametric data.[27]

Poisson regression was used to analyse injury incidence rates (IIRs) with the number of observed injuries as the response variable, cohort period as an independent variable and exposure as an offset. A dispersion test from the 'AER' package[28] was used to test the null hypothesis of equidispersion of the response variable.[29] A quasi-Poisson or negative binomial regression model was used instead where the assumption of equidispersion was not met. The potential confounding effects of age, height, and mass on the response variable was also explored. Overall, match and training IIRs are reported as the number of injuries per 1000 player hours $((\sum \text{injuries} / \sum \text{exposure hours}) \times 1000)$. Injury burden is reported as days absence per 1000 player hours (injury incidence x mean severity). IIRs and rate ratios (RRs) are reported with corresponding 95% CI, with intervals that did not include 1.0 indicating statistical significance. All analyses were conducted using the statistical software R (Version 4.1.0).[30]

Results

A comparison of the control and intervention periods, inclusive of HSI and time-loss groin IIRs and burdens, is shown in Table 1, while an injury burden matrix for HSI and time-loss groin injury is presented in Figure 2. There were no significant differences in the age (p = 0.694), height (p = 0.818), mass (p = 0.613), weekly training sessions (p = 0.162) or weekly matches (p = 0.304) between the intervention and control periods. Overall, there were 10 HSI (6 dominant and 4 non-dominant) and 9

time-loss groin injuries (6 dominant and 3 non-dominant) in the control group, while there were 18 HSI (5 dominant and 13 non-dominant) and 8 time-loss groin injuries (3 dominant and 5 non-dominant) in the intervention group. No adverse events or injuries occurred as a result of strength testing in either the control or intervention periods.

Hamstring Strain Injury

There was a 4.98 days absence/1000 h (19.8%) decrease in HSI burden (intervention period: 20.20 days absence/1000 h; control period: 25.18 days absence/1000 h) in the intervention period compared with the control period. This was largely facilitated via the match-play related burden of HSI more than halving from the control period to the intervention period (48.46 days absence/1000 h (59.0%) decrease). The HSI incidence rates of the control and intervention periods were similar (RR: 1.04; 95% CI [0.49, 2.33]; p = 0.93) (Table 1); however, the severity of HSI was greater in the control period (20 ± 11 days lost/HSI) compared to the intervention period (15 ± 8 days lost/HSI), although this was not statistically significant (p = 0.265).

Time-Loss Groin Injury

There was a 3.93 days absence/1000 h (25.6%) reduction in time-loss groin injury burden (intervention period: 11.44 days absence/1000 h; control period: 15.37 days absence/1000 h) in the intervention period compared with the control period. This was most pronounced in the burden resulting from match-play, with an 18.26 days absence/1000 h (46.7%) decrease in the intervention period compared to the control period. There was a 49% decrease in the incidence of time-loss groin injuries in the intervention period (IIR: 0.59 injuries/1000 h; 95% CI [0.27, 1.11]) compared with that of the control period (IIR: 1.16 injuries/1000 h; 95% CI [0.56, 2.10]), however this did not reach statistical significance (RR: 0.51; 95% CI [0.19, 1.34]; p = 0.17). Time-loss groin injuries were more severe in the intervention period however (19 ± 8 days lost/time-loss groin injury), compared to the control group (13 ± 8 days lost/time-loss groin injury), although this was not statistically significant (p = 0.167).

Alerts

During the intervention period (2018/19 – 2020/21 seasons), there were 30 isometric strength reduction alerts relating to 17 players (33% of the intervention period). Nine arose from right-sided knee flexion (30%), seven from left-sided knee flexion (23%), ten from right-sided hip adduction (33%) and four from left-sided hip adduction (13%) tests. Dominant limbs (57%) were more commonly affected than non-dominant limbs (43%), while forwards (36.5%) were the position most prone to alerts, followed by wide midfielders (20%), wide defenders (13.5%), central defenders (13.5%), goalkeepers (10%) and central midfielders (6.5%).

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Table 1: Comparison of the control and intervention periods

	Control Period (n = 39 players) (n = 2 seasons)	Intervention Period (n = 51 players) (n = 3 seasons)	P value	Effect Size
Age (years) [a]	24.74 ± 5.05	24.32 ± 5.08	0.694	*g* = 0.08
Height (cm) [a]	182.13 ± 6.75	181.79 ± 6.81	0.818	*g* = 0.05
Mass (kg) [a]	78.54 ± 7.35	77.74 ± 7.44	0.613	*g* = 0.11
Weekly training sessions [b]	4 (2, 1 – 5)	4 (2, 1 – 5)	0.162	*r* = 0.15
Weekly matches [b]	1 (0, 1 – 2)	1 (0, 1 – 2)	0.304	*r* = 0.07
Hamstring Strain Injury Number [c]	10 (5)	18 (6)	-	-
Hamstring Strain Injury Incidence [d]	1.29 [0.65, 2.26]	1.34 [0.81, 2.05]	0.930	-
Match	4.01 [1.83, 7.46]	2.31 [0.99, 4.48]	0.289	-
Training	0.35 [0.06, 1.07]	1.05 [0.55, 1.80]	0.150	-
Hamstring Strain Injury Burden [e]	25.18 [13.55, 46.80]	20.20 [12.73, 32.06]	-	-
Match	82.19 [41.10, 164.35]	33.73 [16.08, 70.75]	-	-
Training	5.39 [1.35, 21.55]	16.28 [9.02, 29.40]	-	-
Average days lost/Hamstring Strain Injury [a]	20 ± 11	15 ± 8	0.265	*r* = 0.27
Time-Loss Groin Injury Number [c]	9 (4.5)	8 (2.67)	-	-
Time-Loss Groin Injury Incidence [d]	1.16 [0.56, 2.10]	0.59 [0.27, 1.11]	0.167	-
Match	3.01 [1.20, 6.09]	1.32 [0.41, 3.07]	0.203	-
Training	0.52 [0.13, 1.35]	0.38 [0.12, 0.89]	0.685	-
Time-Loss Groin Injury Burden [e]	15.37 [8.00, 29.54]	11.44 [5.72, 22.88]	-	-
Match	39.09 [17.56, 87.01]	20.83 [7.82, 55.50]	-	-
Training	7.13 [2.30, 22.11]	8.72 [3.27, 23.23]	-	-
Average days lost/Time-Loss Groin Injury [a]	13 ± 8	19 ± 8	0.167	*r* = 0.42

[a] Reported as mean ± standard deviation ($\bar{x}$ ± SD).
[b] Reported as median and (interquartile range (IQR), range)
[c] Reported as total count and (average per season)
[d] Injury incidence reported as the number of injuries per 1000 player hours with corresponding 95% CI
[e] Injury burden reported as days absence per 1000 player hours (injury incidence x mean severity) with corresponding 95% CI.
g = Hedge's *g*, *r* = rank-biserial correlation *r*

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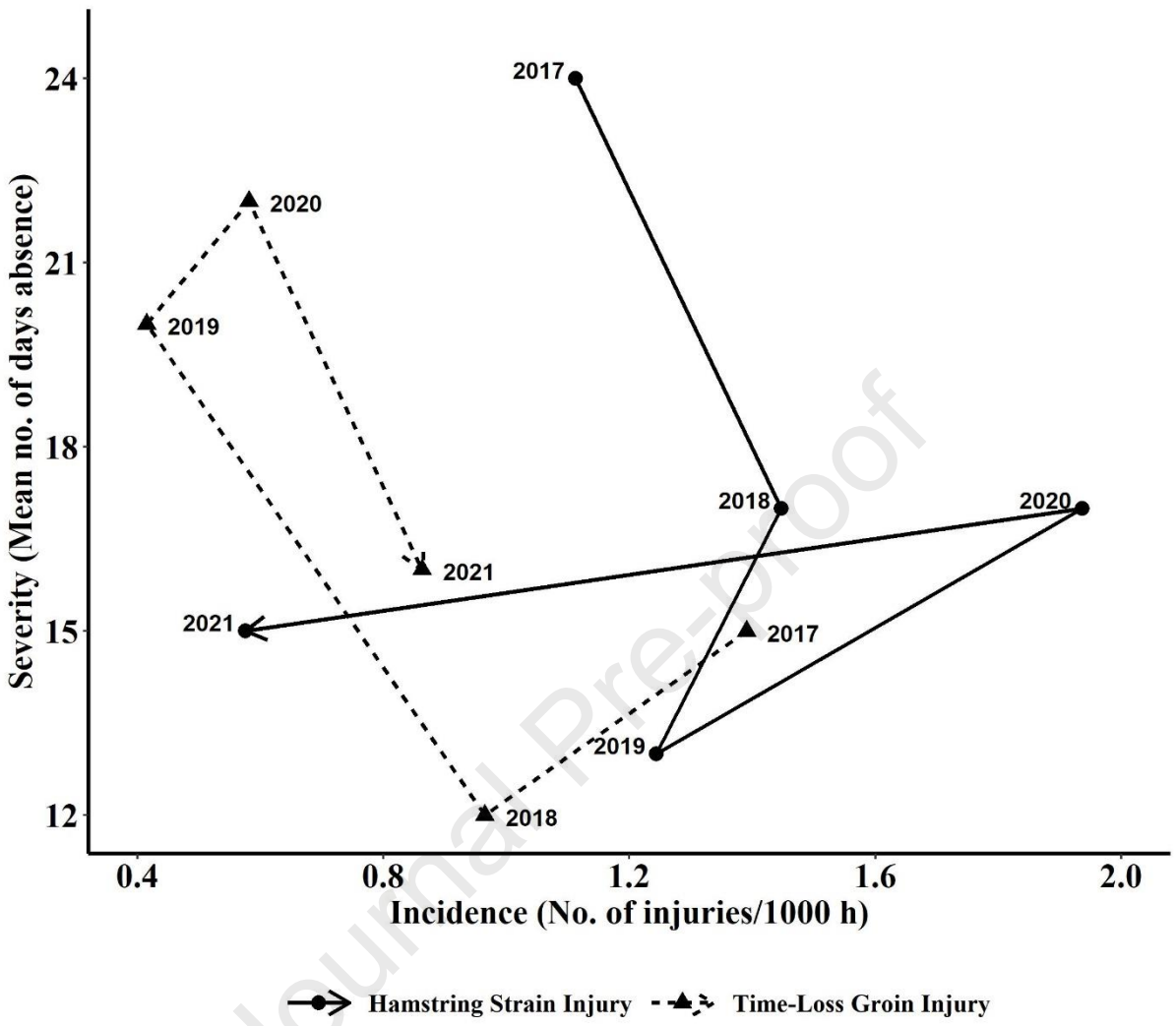

Figure 2: Evaluation of injury burden over five years within a male professional football club for HSI and time-loss groin injuries using a risk matrix (control period – seasons 2017, 2018; intervention period – seasons 2019, 2020, 2021)

Injury incidence reported as the number of injuries per 1000 player hours. Injury severity is defined as the mean number of days absence following injuries. Each point on the graph represents injury burden. Injury burden is reported as days absence per 1000 player hours (injury incidence x mean severity).

Discussion

The aim of this study was to assess the effectiveness of a secondary injury prevention intervention in a systems-based approach for the early detection and management of HSI and time-loss groin injuries within an Australian male professional football club. A key finding of this study was that HSI and time-loss groin injury burdens were reduced during the intervention period when compared with the control period. Following the detection of subclinical forms of injury via greater screening and monitoring practices, a cascade of indicated interventions shows preliminary positive evidence in reducing the severity of HSI, albeit non-significantly. Forwards and wide positions within the intervention period were more prone to post-match isometric strength reduction alerts compared with central positions. This highlights the potential for individual and position-based modifications to subsequent training sessions in the post-match period as part of a secondary injury prevention system.

Adhering to the system parameters outlined in Figure 1, a reduction in the burdens of both HSI and time-loss groin injury in male professional football players was observed, replicating results in the literature of similar methodology.[6] A notable difference between our current study and the previous study is the level of competition and the instability of coaching staff during these periods. Previously, a novel HSI prevention system incorporating secondary prevention in addition to usual care (primary and tertiary prevention) was able to lower HSI burden in youth football players (12.9 days absence/1000 h (90.8%) decrease) when compared to usual care only (intervention period: 1.3 days absence/1000 h; control period: 14.2 days absence/1000 h).[6] While similarly, an early detection and management strategy for the monitoring of hip and groin strength was able to reduce in-season hip and groin injury burden when compared to pre-season (mean difference: -5.97 days absence; 95% CI [-11.27, -0.66]).[7] It must be noted both studies were completed in youth football thus limiting the applicability to our current study. Nevertheless, the implementation of early detection and indicated interventions alongside primary and tertiary injury prevention shows promise in reducing HSI and time-loss groin injury burdens.

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We identified post-match isometric strength reduction trends which may be of value for practitioners in the context of post-match monitoring and subsequent training prescription. Dominant limbs in our study were more likely to have strength reductions than non-dominant limbs during testing two days post-match (≥40 h). This is consistent with previous research in which post-match non-dominant isometric hamstring strength (90° knee flexion) recovers quicker than that of dominant, with suggested links to a higher frequency of technical actions (long passes and shots) performed using the dominant limb.[31] We also observed over a third of all strength reduction alerts to involve 'forwards', while wide positions (wide midfielders and wide defenders) were more prone to alerts than central positions (central defenders and central midfielders). This may be attributed to forward and wide positions completing a greater number of sprints and total sprint distance ($p < 0.001$) during match-play compared with central positions.[22] Together these findings provide practitioners with relevant information to inform individual and position-based modifications to subsequent training sessions in the post-match period as part of a secondary injury prevention system.

While the results of the present study suggest the introduction of secondary injury prevention within male professional football may be useful to clubs, we refrain from advocating their use in isolation. Recent study has shown repeated strength testing as a means of monitoring has limited predictive value for HSI,[32] while there is limited evidence to support primary prevention exercise-based strategies alone in preventing muscle injury in elite football.[5] Instead, complementing the various pathology-based (primary, secondary, and tertiary prevention) and risk factor-based (universal, selective and indicated) injury prevention categorisations via an integrated systems-based approach is recommended.[6,7,33] Despite these recommendations, there was no statistically significant evidence to suggest the daily bilateral hip adductor squeeze peak force and hamstring (sit-and-reach) flexibility testing in the control period failed to provide valuable insights to assist in reducing injury burden when compared to in-season post-match testing within a secondary prevention systems-based approach.

A strength of this study is that it was undertaken within a male professional football club over multiple seasons. The real-world application of the utilised injury prevention framework suggests its

implementation elsewhere is realistic and feasible. The test protocols are quick to administer, acceptable to players and coaching staff and may be implemented in-season without interference to team schedules or causing injury. Interventions are individualised and may be particularly pertinent during congested match schedules. However, there are limitations to be noted within this study. We saw an increase, albeit not statistically significant, in HSI incidence in the intervention period compared with the control period. This increase in incidence may be due to a concept termed the 'lead time problem' where the practice of increased screening and monitoring practices leads to a greater number of cases being identified.[8] Despite this increase in incidence, the early identification of injuries enabled the expedited application of indicated interventions to reduce injury severity and burden and to subsequently increase player availability, which is linked to positive performance outcomes in football.[34]

Due to the nature of professional sport, staffing changes across the study period were inevitable. There were four head coach, two physiotherapist and two strength and conditioning coach changes across our five-season study period. Due to this instability, internal relationships and communication quality between the coaching staff may have been affected. This is particularly relevant given high communication quality between the medical team and head coach is associated with low injury burden, low incidence of severe injuries, high training attendance and high match availability.[35] Similarly, different coaching leadership styles have been proven to affect injury rates and player availability, with a negative correlation between the overall level of transformational leadership of the head coach and the incidence of severe injuries.[36] While attributing causation is beyond the scope of our study, the effect of coaching and medical staff instability is unknown, yet is likely to have contributed to the difference in the outcome during periods of transition.[35,36] As staffing changes are frequent in professional football, this further supports the need for solid monitoring systems to help mitigate the influence of changing staff profiles on player injury incidence and burden rates.

Results must also be considered with the impact of the COVID-19 pandemic in mind. There was an increase in training injury burdens for HSI and time-loss groin injuries during the intervention period (encompassing the restart to the 2019/2020 season following a three-month cessation of all team-based

activity due to COVID-19) compared with the control period. Recent research supports this trend, with spikes identified in training injury burdens[37] with links to short-term detraining effects due to quarantine/periods of inactivity.[38] Similarly, due to season schedule changes stemming from COVID-19, there was a reduced pre-season for the 2020/21 season, with 65 days available for pre-season training compared with a mean of 109 days from the preceding four seasons. This is particularly pertinent given an increase of 10 pre-season sessions is linked with a decrease in severe injury incidence (-0.20 severe injuries/1000 h) and injury burden (-21.8 layoff days/1000 h).[39] Despite an increase in training injury burdens linked to the COVID-19 pandemic, overall HSI and time-loss groin injury burdens were still reduced in the intervention period when compared with the control period.

Finally, a non-randomised study design reduces the strength of the results obtained in the present study, which must be considered when interpreting the practical implications of this research. Despite a five-season monitoring period, the relative strength of the study was also restricted by a sample size limited to 21 – 31 players per season pertaining to one club only. The ability to generalise results outside of this specific context is therefore limited. A risk of confirmation bias may also have been present within this study. Given the distinction between end of time loss and return to play is not objectively clear, [40] the possibility of sports medicine staff subconsciously influencing return to play decision making of injured players to improve the effectiveness of the intervention must also be considered.

Conclusion

The implementation of a systems-based secondary injury prevention intervention provides preliminary positive findings in reducing HSI and time-loss groin injury burdens in a cohort of male professional football players, when compared with a control period. This study provides practitioners with a practical framework that can be applied in male professional football, that is acceptable to players and coaches, without interfering with normal team schedules.

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Declaration of Interests

No potential conflict of interest was reported by the author(s).

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Highlights

- Maximum voluntary isometric contractions may be safely monitored post-match play.
- Indicated interventions may be individual- and position-specific.
- Systems-based prevention may reduce hamstring and time-loss groin injury burden.

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Ethical Approval

Ethical approval was obtained from the University of Newcastle, Australia (reference number H-2018-0229) and all players provided written consent to participate in the study.

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